

# TASK 1: DNA/Protein Sequence Analysis Using BLAST

## Title

BLAST Analysis of Human Hemoglobin Beta Protein (HBB)

## Objective

To identify homologous protein sequences of Human Hemoglobin Beta Protein using BLAST and analyze sequence similarity based on alignment score, query coverage, identity percentage, and E-value.

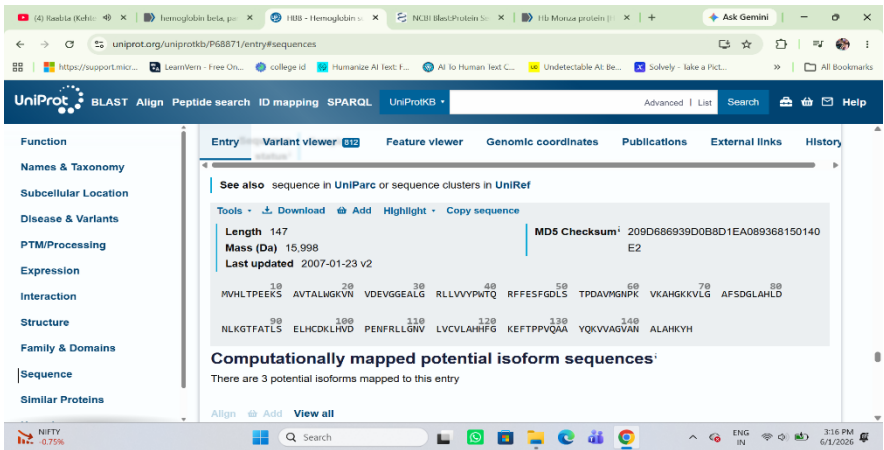
## Introduction

BLAST (Basic Local Alignment Search Tool) is a bioinformatics tool used to compare biological sequences and identify similar sequences in databases. In this experiment, the Human Hemoglobin Beta Protein sequence was obtained from UniProt and analyzed using NCBI BLASTP to identify homologous proteins.

## Protein Information

| Parameter       | Details                 |
|-----------------|-------------------------|
| Protein Name    | Hemoglobin Subunit Beta |
| Gene Name       | HBB                     |
| Organism        | Homo sapiens            |
| UniProt ID      | P68871                  |
| Sequence Length | 147 Amino Acids         |

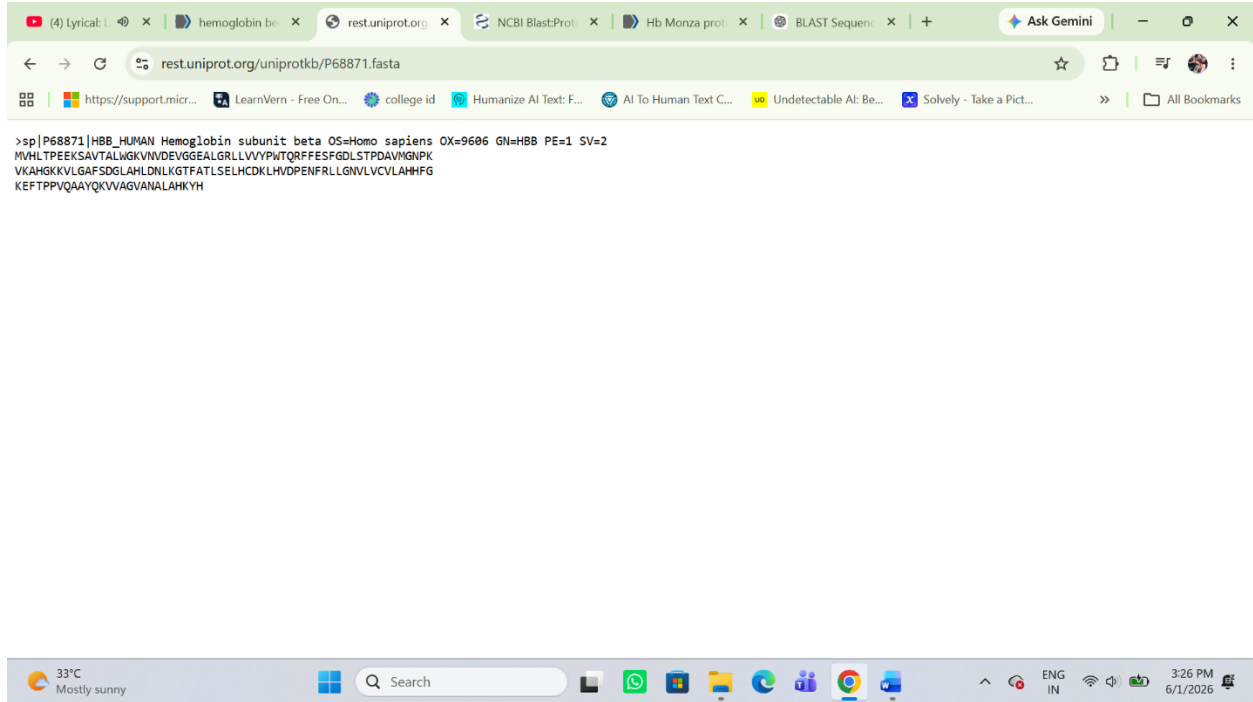
## Screenshot 1: UniProt Protein Entry



## FASTA Sequence

The protein sequence was downloaded in FASTA format from UniProt and used as the query sequence for BLAST analysis.

### Screenshot 2: FASTA Sequence



## Methodology

1. Human Hemoglobin Beta Protein sequence was downloaded from UniProt.
2. The FASTA sequence was copied.
3. NCBI BLASTP was opened.
4. The FASTA sequence was pasted into the query box.
5. Default BLAST parameters were used.
6. BLAST search was executed.
7. Homologous protein sequences were analyzed.
8. Query coverage, identity percentage, E-value, and alignment score were recorded.

# BLAST Results

## Screenshot 3: BLAST Results Page (Clusters)

BLAST® » blastp suite » results for RID-1UMPSSAX014

Home Recent Results Saved Strategies Help

[Edit Search](#) [Save Search](#) [Search Summary](#) [How to read this report?](#) [BLAST Help Videos](#) [Back to Traditional Results Page](#)

**Job Title** Protein Sequence

**RID** 1UMPSSAX014 [Search expires on 06-02 17:37 pm](#) [Download All](#)

**Program** BLASTP [Citation](#)

**Database** ClusteredNR [See details](#)

**Query ID** lcl|Query\_4400876

**Description** unnamed protein product

**Molecule type** amino acid

**Query Length** 147

**Other reports** [Distance tree of results](#) [Multiple alignment](#) [MSA viewer](#)

**Filter Results**

**Organism** only top 20 will appear **NEW**

Type common name, binomial, taxid or group name

[Add organism](#)

**Percent Identity**  to  **E value**  to  **Query Coverage**  to

[Filter](#) [Reset](#)

**Clusters** Graphic Summary Alignments Taxonomy

**Clusters producing significant alignments** [Download](#) [Select columns](#) Show 100 [?](#)

[select all](#) 100 clusters selected

[GenPept](#) [Graphics](#) [Distance tree of results](#) [Multiple alignment](#) [MSA Viewer](#)

**Clusters** Graphic Summary Alignments Taxonomy

**Clusters producing significant alignments** [Download](#) [Select columns](#) Show 100 [?](#)

[select all](#) 100 clusters selected

[GenPept](#) [Graphics](#) [Distance tree of results](#) [Multiple alignment](#) [MSA Viewer](#)

|                                     | Cluster Composition                        | Cluster Ancestor    | Cluster Representative Sequence                                                       | Max Score | Total Score | Query Cover | E value | Per. Ident | Acc. Len | Accession      |
|-------------------------------------|--------------------------------------------|---------------------|---------------------------------------------------------------------------------------|-----------|-------------|-------------|---------|------------|----------|----------------|
| <input checked="" type="checkbox"/> | Click the icon to see the cluster contents |                     |                                                                                       |           |             |             |         |            |          |                |
| <input checked="" type="checkbox"/> | 1 member(s), 1 organism(s)                 | human               | Hb Monza protein [Homo sapiens]                                                       | 288       | 288         | 100%        | 1e-97   | 86.47%     | 170      | XE57200.1      |
| <input checked="" type="checkbox"/> | 1 member(s), 1 organism(s)                 | human               | beta-globin [Homo sapiens]                                                            | 277       | 277         | 100%        | 2e-93   | 93.20%     | 147      | AAA35597.1     |
| <input checked="" type="checkbox"/> | 7 member(s), 6 organism(s)                 | dog_coyote_wolf_fox | hemoglobin subunit beta-like [Canis lupus familiaris]                                 | 275       | 275         | 100%        | 1e-92   | 89.80%     | 147      | NP_001257812.1 |
| <input checked="" type="checkbox"/> | 183 member(s), 75 organism(s)              | placentals          | hemoglobin subunit delta [Ailuropoda melanoleuca]                                     | 273       | 273         | 100%        | 6e-92   | 89.80%     | 147      | NP_001291814.1 |
| <input checked="" type="checkbox"/> | 19 member(s), 18 organism(s)               | placentals          | hemoglobin subunit beta-like [Ailuropoda melanoleuca]                                 | 273       | 273         | 100%        | 7e-92   | 89.80%     | 147      | NP_001291812.1 |
| <input checked="" type="checkbox"/> | 7 member(s), 5 organism(s)                 | bats                | hemoglobin subunit beta [Rhynchonycteris naso]                                        | 271       | 271         | 100%        | 2e-91   | 89.12%     | 147      | KAM8815943.1   |
| <input checked="" type="checkbox"/> | 1 member(s), 1 organism(s)                 | giant_panda         | hypothetical protein PANDA_015769 [Ailuropoda melanoleuca]                            | 275       | 446         | 100%        | 3e-91   | 90.48%     | 258      | EFB18430.1     |
| <input checked="" type="checkbox"/> | 1 member(s), 1 organism(s)                 | large_flying_fox    | TPA: globin A2 [Pteropus vampyrus]                                                    | 272       | 272         | 100%        | 7e-91   | 88.44%     | 193      | SAI82279.1     |
| <input checked="" type="checkbox"/> | 4 member(s), 4 organism(s)                 | primates            | PREDICTED: LOW QUALITY PROTEIN: hemoglobin subunit beta-like [Ailuropoda melanoleuca] | 270       | 270         | 100%        | 8e-91   | 89.80%     | 147      | XP_011830554.1 |
| <input checked="" type="checkbox"/> | 38 member(s), 26 organism(s)               | cellular organisms  | hemoglobin beta-like protein, partial [Apibacter sp. B3889]                           | 269       | 269         | 100%        | 3e-90   | 87.76%     | 162      | WP_221411036.1 |
| <input checked="" type="checkbox"/> | 20 member(s), 12 organism(s)               | carnivores          | hemoglobin subunit beta isoform X3 [Neomonachus schauinslandi]                        | 267       | 267         | 100%        | 1e-89   | 86.39%     | 147      | XP_044775197.1 |

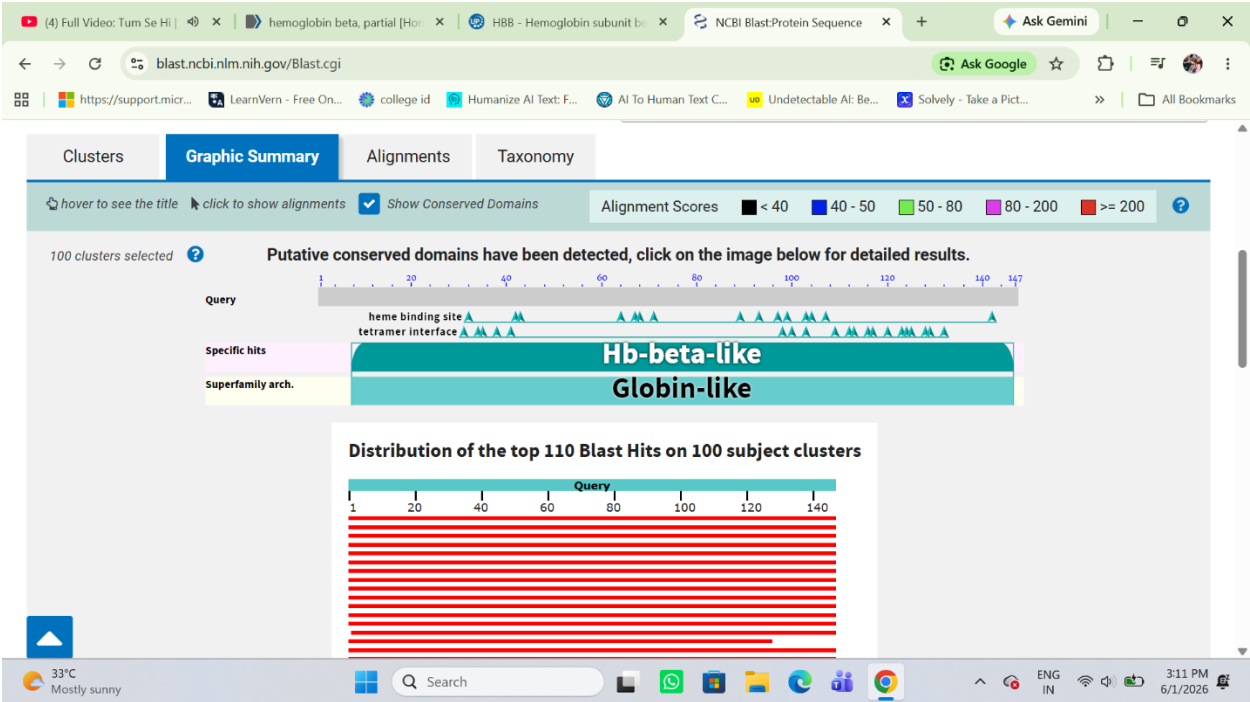
BLAST Result Summary

| Parameter        | Value                           |
|------------------|---------------------------------|
| Query Length     | 147 aa                          |
| Query Coverage   | 100%                            |
| Percent Identity | 86%                             |
| E-value          | 1e-97                           |
| Alignment Score  | 288 bits                        |
| Top Match        | Hb Monza Protein (Homo sapiens) |

Graphic Summary

The graphic summary showed significant alignments between the query sequence and homologous protein sequences. Conserved globin domains and heme-binding regions were detected.

Screenshot 4: Graphic Summary



## Alignment Analysis

The pairwise alignment showed a high degree of similarity between the query sequence and the matched protein sequence. The low E-value indicates that the alignment is statistically significant and not due to random chance.

## Screenshot 5: Alignment Results

The screenshot displays the NCBI Blast results page for a protein sequence alignment. The interface includes tabs for Clusters, Graphic Summary, Alignments (selected), and Taxonomy. The alignment view is set to Pairwise. The results show 100 clusters selected.

**Hb Monza protein [Homo sapiens]**  
Sequence ID: [XEF57200.1](#) Length: 170 Number of Matches: 1

Range 1: 1 to 170 [GenPept](#) [Graphics](#) [Next Match](#) [Previous Match](#)

| Score         | Expect | Method                       | Identities   | Positives    | Gaps        |
|---------------|--------|------------------------------|--------------|--------------|-------------|
| 288 bits(737) | 1e-97  | Compositional matrix adjust. | 147/170(86%) | 147/170(86%) | 23/170(13%) |

Query 1 MVHLTPEEKSAVTALWGKVNVDGGEALGRLLVVPWTRQFFESFGDLSTPDAMGNPK 60  
Sbjct 1 MVHLTPEEKSAVTALWGKVNVDGGEALGRLLVVPWTRQFFESFGDLSTPDAMGNPK 60

Query 61 VKAHGKKVLGAFSDGLAHLDN-----LKGTFATLSELHCDKL 97  
Sbjct 61 VKAHGKKVLGAFSDGLAHLDNPKVKAHGKKVLGAFSDGLAHLDNLKGTFATLSELHCDKL 120

Query 98 HVDPENFRLLGNVLCVLAHHFGKEFTPPVQAAQYQKVAVANALAHKYH 147  
Sbjct 121 HVDPENFRLLGNVLCVLAHHFGKEFTPPVQAAQYQKVAVANALAHKYH 170

**beta-globin [Homo sapiens]**  
Sequence ID: [AAA35597.1](#) Length: 147 Number of Matches: 1

Range 1: 1 to 147 [GenPept](#) [Graphics](#) [Next Match](#) [Previous Match](#)

| Score         | Expect | Method                       | Identities   | Positives    | Gaps      |
|---------------|--------|------------------------------|--------------|--------------|-----------|
| 277 bits(708) | 2e-93  | Compositional matrix adjust. | 137/147(93%) | 137/147(93%) | 0/147(0%) |

Query 1 MVHLTPEEKSAVTALWGKVNVDGGEALGRLLVVPWTRQFFESFGDLSTPDAMGNPK 60  
Sbjct 1 MVHXTPEKSAVTALWGKVNVDGGEALGXLLVVPWTRQFFESFGDLSTPDAMGNPK 60

Query 61 VKAHGKKVLGAFSDGLAHLDNLKGTFATLSELHCDKLHVDPENFRLLGNVLCVLAHHFG 120  
Sbjct 61 VKAHGKKVLGAFSDGLAHLDNLKGTFATLSELHCDKLHVDPENFRLLGNVLCVLAHHFG 120

Query 121 KEFTPPVQAAQYQKVAVANALAHKYH 147  
Sbjct 121 KEFTPPVQAAQYQKVAVANALAHKYH 147

**hemoglobin subunit beta-like [Canis lupus familiaris]**  
Sequence ID: [NP\\_001257812.1](#) Length: 147 Number of Matches: 1

Related Information  
[Gene](#) - associated gene details

## Alignment Statistics

### Parameter Value

|            |               |
|------------|---------------|
| Score      | 288 bits      |
| E-value    | 1e-97         |
| Identities | 147/170 (86%) |
| Positives  | 147/170 (86%) |
| Gaps       | 23/170 (13%)  |

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## Discussion

The BLASTP analysis identified homologous hemoglobin proteins with high sequence similarity. The query coverage of 100% indicates that the complete protein sequence was aligned with the subject sequence. An identity value of 86% suggests strong evolutionary conservation among globin proteins. The extremely low E-value (1e-97) confirms that the observed similarity is highly significant. Conserved globin domains and heme-binding sites indicate the functional importance of the protein.

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## Conclusion

BLAST analysis successfully identified homologous protein sequences related to Human Hemoglobin Beta Protein. The results demonstrated high sequence similarity, complete query coverage, and significant alignment scores, confirming the conserved nature of hemoglobin proteins among related organisms.

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## References

1. UniProt Protein Database.
2. NCBI BLAST (Basic Local Alignment Search Tool).
3. National Center for Biotechnology Information (NCBI).